CHAPTER 1

INTRODUCTION

1.1 Stomata, water and global food security

Stomata are microscopic pores on the leaf epidermis that regulate the exchange of water vapor (H_2O) and carbon dioxide (CO_2) (Fig. 1.1) between terrestrial vegetation and the atmosphere, accounting for more than 95% of all terrestrial gaseous flux of water and carbon[1]. Agriculture currently accounts for roughly 70% of all freshwater withdrawals worldwide, and climate projections consistently indicate that drought frequency and severity will increase through the twenty-first century[2, 3, 4]. Improving crop water-use efficiency (WUE), defined as the ratio of carbon assimilation (A [$\mu mol\ m^{-2}\ s^{-1}$]) to transpiration water loss (E [$mmol\ m^{-2}\ s^{-1}$]), is therefore a central challenge for food security under a warming and water-limited climate.

Stomatal aperture governs the critical trade-off between CO_2 uptake for photosynthesis and water loss via transpiration. Both fluxes are regulated by the stomatal resistance ($1/g_s$), making stomatal conductance, g_s [$mmol\ m^{-2}\ s^{-1}$], a crucial determinant of plant productivity, WUE, and drought resilience[5, 6]. In the soil plant atmosphere continuum (SPAC), water potential (ψ [MPa]) decreases from the soil (ψ_{soil}) through the xylem (ψ_{xyl}) and the leaf apoplast at the evaporating surface of the sub-stomatal cavity (ψ_{ssc}^{apo}) to the atmosphere (ψ_{air}), driving water flow while the stomatal resistance sets the A–E trade-off at the pore, a “gamble” in Graham Farquhar’s words[7].

Despite decades of research, stomatal physiology still lacks a unified mechanistic account of how water status, hydraulic transport, and biochemical signaling are integrated during water stress. In particular, existing frameworks have not fully resolved how hydraulic architecture at the leaf-scale, guard-cell ABA biosynthesis and signaling, and outside-xylem resistance interact to determine stomatal behavior across species and environments[8, 9]. This gap limits the ability to explain variation in stomatal responses and constrains efforts to predict or improve crop water-use efficiency under increasing water limitation[10, 4]. A clearer mechanistic understanding of this regulation is therefore needed to connect plant water relations to crop performance and global food security[11].

1.2 The soil plant atmosphere continuum (SPAC)

Water movement through the SPAC is driven by gradients in water potential (ψ [MPa]), which decrease from the soil through the xylem, outside-xylem zone (OXZ) to the sub-stomatal cavity apoplast and then to the atmosphere. Water potential in the SPAC spans an immense range, from near zero values of roughly -0.001 MPa in moist soils to extremely negative values approaching -100 MPa in dry air[12]. Each of these compartments represents a hydraulic node connected by resistances and capacitances that together shape leaf water status. Cowan [13] formalized this perspective by treating the SPAC as an electrical RC network in which water potential is analogous to voltage and water flux is analogous to current. The plant leaf operates between two boundary conditions: water availability, quantified by xylem water potential ψ_{xyl}, and atmospheric water demand, quantified by the vapor pressure deficit (VPD [kPa]). Jarvis [14] first systematically characterized how stomatal conductance g_s responds as a

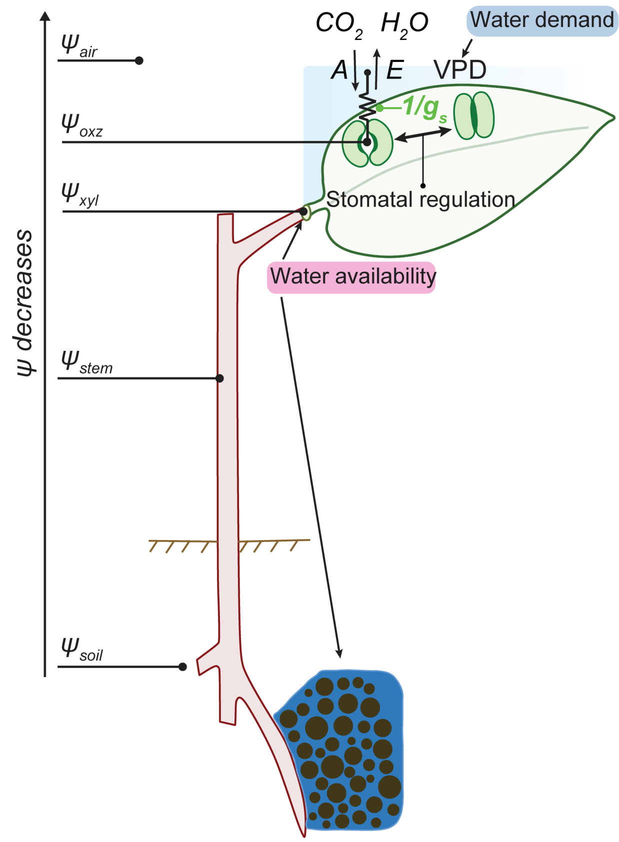

Figure 1.1. Schematic representation of water transport through the soil-plant-atmosphere continuum (SPAC). Water flows passively along a thermodynamic gradient, driven by decreasing water potential ψ $[MPa]$ from the root zone (ψ_{soil}) through the plant vascular system (ψ_{stem}, ψ_{xyl}) to the leaf tissue (ψ_{ssc}^{apo}) and ultimately the atmosphere (ψ_{air}). The plant operates under two primary boundary conditions: water availability, reflected by xylem water potential (ψ_{xyl}), and atmospheric water demand, governed by the vapor pressure deficit (VPD $[kPa]$). At the leaf level, stomatal regulation acts as a variable hydraulic resistance (where g_s $[mmol/m^2/s]$ is stomatal conductance), modulating the vital trade-off between the efflux of transpired water vapor (H_2O) E $[mmol/m^2/s]$ and the influx of CO_2 for photosynthetic assimilation A $[\mu mol/m^2/s]$.

multiplicative function of light, VPD, temperature, CO_2, and leaf water potential. This empirical framework is practically useful at canopy and land-surface scales but treats stomata as a black box, without resolving the mechanistic links between environmental drivers, tissue hydraulics, and guard cell biophysics that motivate this thesis.

At the stomatal pore, increasing g_s raises both carbon assimilation A and transpirational water loss E, and the intrinsic water-use efficiency (A/g_s) captures this trade-off in a way that is convenient for modelling[15]. However, finite outside-xylem (OXZ) resistance means that stomata are not always the sole limiting pathway for water vapor, a fact that standard gas-exchange analyses can obscure because they assume saturated intercellular air spaces[16, 17, 18].

The soil represents the least-negative node in the hydraulic continuum[12, 19, 20]. As soil dries, soil water potential ψ_{soil} becomes more negative and the unsaturated hydraulic conductivity declines by several orders of magnitude as pores empty and the connected liquid network breaks down[21, 22]. Neutron-radiography measurements now show that hydraulic limitation often arises first at the root–soil interface rather than in the xylem or leaf, underscoring soil water dynamics in the rhizosphere as a critical component of plant drought response[23, 24, 25].

1.3 Leaf anatomy, stomatal complex and emergent behaviors of stomatal conductance

A leaf cross-section, illustrated in Fig. 1.2a, reveals the pathway that the transpiration stream must traverse: from vascular tissue (xylem), through bundle sheath and mesophyll cells, to subsidiary and guard cells in the leaf epidermis, before water evaporates from cell interfaces into the sub-stomatal cavity and exits the leaf through the stomatal pore. The leaf is geometrically divided into an adaxial surface above the xylem plane and an abaxial surface below it, and in C_4 species such as maize the bundle sheath is hydraulically coupled to the mesophyll. In the epidermis, illustrated in Fig. 1.2b–c, stomata are formed by pairs of guard cells flanked by subsidiary cells[26, 27, 28].

The mechanical interaction between the subsidiary cells and guard cells in the stomatal complex is critical for stomatal behavior. Subsidiary cells exert a backpressure on the guard cells, so that a uniform decline in leaf water potential can open the pore because subsidiary cell turgor relaxes faster than guard cell turgor[29]. This "mechanical advantage" of the subsidiary cells relative to guard cells underlies hydropassive responses and provides a structural basis for counter-intuitive stomatal dynamics.

Between the xylem and the guard cells lies the outside-xylem zone (OXZ), comprising mesophyll symplasm, apoplast, cell walls, and the intercellular vapor space[18]. Standard gas-exchange analysis following Gaastra [16] assumes that the leaf air space is water-vapor-saturated, which is equivalent to assuming infinite OXZ conductance and assigning all resistance to the stomatal pore. This assumption was challenged first by Jarvis and Slatyer [30] and is definitively disproven by AquaDust measurements, which show large drops in water potential between xylem and sub-stomatal cavity apoplast[31, 32, 33]. The ratio g_s/g_{oxz} therefore quantifies the relative importance of stomatal versus non-stomatal resistance and serves as a mechanistically grounded phenotype for water-use efficiency.

Guard cells provide an active biochemical response to water stress through the stress hormone abscisic acid (ABA) signaling under water stress, which regulates their osmotic content and turgor. Under declining water status, ABA accumulates in guard cells and triggers ion efflux, reducing osmotic potential and closing the pore[34, 35, 36, 37, 38]. Stomatal behavior under water stress thus emerges from the interaction between hydropassive (HP) mechanical advantage and hydroactive (HA) ABA-driven control of guard cell osmotic potential. Together, these HP and HA processes generate the characteristic transient

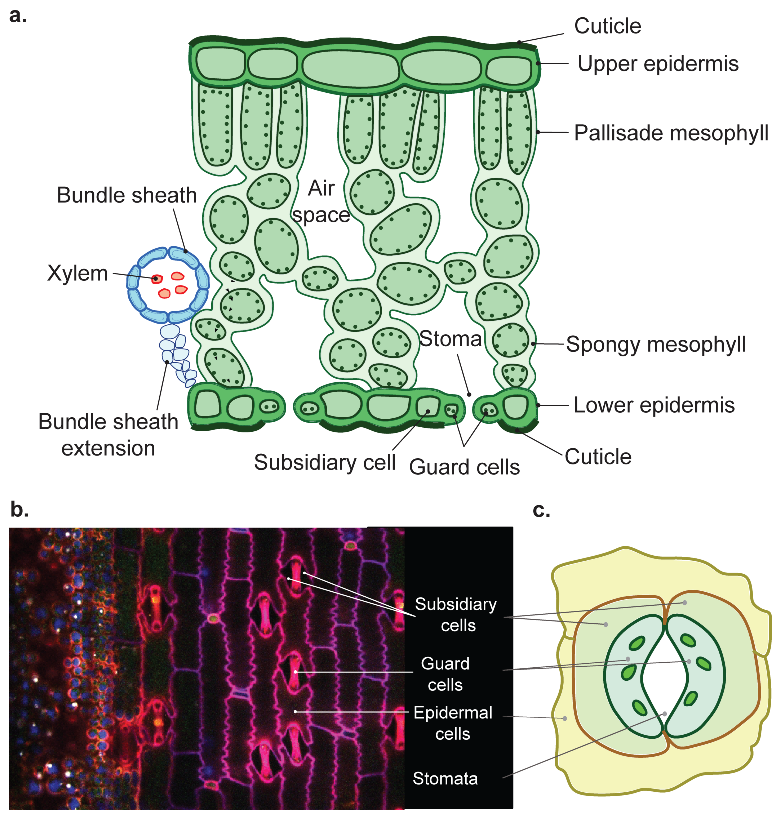

Figure 1.2. Leaf anatomy and the architecture of the stomatal complex. a. Schematic diagram of the cross-section of a leaf showing the xylem, mesophyll and sub-stomatal cavity in the outside xylem zone of leaf tissue. The sub-stomatal cavity comprises a variety of different cells that facilitate the flow of water and the adsorption of CO_2. Xylem is the main vascular vessel that carries water from the roots to the leaves. The leaf is divided into two regions: adaxial, the surface above the plane of xylem, and abaxial, the surface below the plane of xylem. Water moves from xylem to bundle sheath, mesophyll, subsidiary, and guard cells; it evaporates from the interfaces of the mesophyll cells within the internal vapor spaces before it is lost to the atmosphere as transpiration stream. **b.** Confocal microscopy image of maize epidermis. Line segments indicate the positions of representative guard cells and subsidiary cells within the epidermis. The red fluorescence spots among the pink fluorescence in the guard cell are due to chloroplast autofluorescence. Such autofluorescence is not evident in other epidermal cells because they lack chloroplasts. *Image courtesy: Piyush Jain.* **c.** Schematic representation of epidermal, subsidiary, guard cells, and the stomata formed by a pair of guard cells.

and steady-state patterns of stomatal conductance illustrated in Fig. 1.3. Fig. 1.3 illustrates the three transient and steady-state behaviors that define the phenomenological targets of this thesis. Upon a step increase in VPD, g_s briefly rises before declining to a new, lower steady state, a transient termed the wrong-way response (WWR). The WWR was first modelled mechanistically by Cowan [13] and Delwiche and Cooke [39] as arising from the phase lag between subsidiary and guard cell turgor responses in a purely HP framework.

To overcome this initial WWR and achieve sustained stomatal closure, guard cells must actively adjust their osmotic potential, producing a right-way response (RWR) that reflects the hydroactive ABA-driven component of regulation. Under moderate step increases in VPD, during closure g_s exhibits damped oscillations before reaching steady state; under high VPD, oscillations can persist indefinitely, as shown in Fig. 1.3a. This transition from damped to sustained oscillations is observed experimentally by Farquhar and Cowan [40] but not yet mechanistically grounded.

Fig. 1.3b illustrates stomatal hysteresis: under cycles of drydown and recovery, g_s follows different paths for the same ψ_{xyl}, so that g_s is not a single-valued function of ψ_{xyl} but depends on the history of water-status trajectory. Experiments have characterized this hysteresis and speculated guard-cell signaling components associated with it, but the mathematical and mechanistic origins of the hysteresis itself remain unresolved[41, 42, 43]. Beyond these dynamic behaviors, steady-state g_s measurements under drought routinely overestimate true stomatal conductance when internal air-space undersaturation occurs[33, 17, 18]. Jain et al. [33] showed in maize that g_s/g_{oxz} can approach 0.3–0.5, leading to 20–25% errors in standard LI-COR gas-exchange calculations, underscoring the need to characterize this non-stomatal OXZ phenotype alongside stomatal regulation itself.

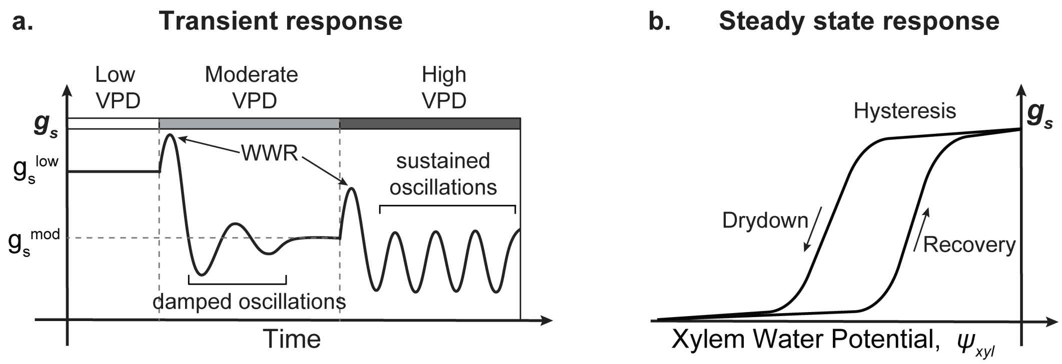

Figure 1.3. Characteristic dynamic and steady-state behaviors of stomatal conductance. a. Transient response of stomatal conductance (g_s) to stepwise increases in atmospheric demand (low → moderate → high vapor pressure deficit, VPD). Each step elicits a transient "wrong-way response" (WWR), in which g_s initially moves opposite to its eventual steady-state direction before relaxing toward a new equilibrium (g_s^{low}, g_s^{mod}). Depending on the magnitude of the forcing, this relaxation manifests as damped oscillations (moderate VPD) or, under high VPD, as sustained oscillations that fail to settle. **b.** Steady-state relationship between g_s and xylem water potential (ψ_{xyl}), which exhibits hysteresis: the drydown trajectory (decreasing ψ_{xyl}) and the recovery trajectory (increasing ψ_{xyl}) follow distinct paths, so that g_s is not a single-valued function of ψ_{xyl} but depends on the history of the water-status trajectory.

1.4 Review of stomatal conductance models

Stomatal models fall into three broad categories that reflect different philosophies about what constitutes a satisfactory explanation of plant behavior: empirical, optimality-based, and mechanistic (process-based)[10, 5, 8]. These categories are not mutually exclusive and address different spatial and temporal scales and purposes, but their cumulative limitations motivate the mechanistic synthesis pursued in this thesis.

1.4.1 Empirical models: fitting the phenotype

Jarvis [14] expressed g_s as a multiplicative product of dimensionless response functions for light, VPD, temperature, CO_2, and leaf water potential, with each function fitted independently from data and no mechanistic coupling between environmental drivers. The model is widely implemented in canopy-scale and land-surface models because of its simplicity, but it cannot predict any of the transient behaviors in Fig. 1.3 because it has no dynamics, and its parameters are species-, site-, and season-specific.

Ball et al. [44] introduced the insight that g_s is empirically proportional to the product of net assimilation (A), relative humidity at the leaf surface, and the inverse of CO_2 concentration at the leaf surface. This coupling to A was a major advance because it explained the observed correlation between g_s and photosynthesis. Ball–Berry remains the most widely implemented stomatal module in global vegetation and climate models, yet it contains no mechanistic water-stress term and is entirely steady-state. Medlyn et al. [10] developed upon the Ball–Berry framework, replacing the empirical RH term with a VPD-based function, providing a more physically interpretable but still steady-state formulation.

Leuning [45] replaced the relative humidity term in the Ball–Berry model with a hyperbolic VPD function and introduced the CO_2 compensation point, making the model more physically interpretable, but the framework remains fundamentally empirical. Empirical models therefore remain effective black boxes: they cannot explain the mechanistic basis of stomatal responses, cannot be reliably extrapolated beyond the environmental and species domains for which they were calibrated, and fail to capture newer findings such as substantial outside-xylem (OXZ) limitation of transpiration revealed by AquaDust measurements in maize[33].

1.4.2 Optimality models: explaining the phenotype from first principles

Cowan and Farquhar [15] posited that stomata behave so as to maximize carbon assimilation over a given period subject to a fixed water budget, yielding the optimality condition $\partial A/\partial E = \lambda$. This theory captures the observed tendency of stomata to maintain a nearly constant balance between internal and ambient CO_2 across diverse environments without fitting species-specific parameters, but it treats λ as an externally specified constant with no direct physical basis, provides no mechanism by which stomata achieve the optimum, and contains no dynamics.

Katul et al. [46] demonstrated analytically that if the marginal water cost scales as atmospheric CO_2 concentration, the optimal g_s displays the CO_2-sensitivity observed empirically, providing a mechanistic rationale for Ball–Berry-type coupling. Medlyn et al. [10] reconciled the optimal and empirical approaches to stomatal modelling, showing that the optimal stomatal theory predicts a specific functional form relating g_s to A, VPD, and ambient CO_2 concentration that is consistent with experimental data. Carminati and Javaux [23] used a hydraulic supply-function model to show that soil physical constraints can drive convergence in the critical stomatal water potential across diverse species, suggesting that drought responses are strongly shaped by soil–plant hydraulics rather than guard-cell biochemistry alone. Despite these insights, optimality-based models remain phenomenological: they encode what stomata should do under given constraints but lack a mechanistic description of how g_s is regulated in real time, and they could therefore benefit from integration with process-based stomatal models that explicitly represent hydraulic and ABA signaling mechanisms[8, 9].

1.4.3 Mechanistic (process-based) models: resolving the how

Cowan's [13] landmark paper established both the RC-network representation of the SPAC, setting up a theoretical basis for the hydropassive (HP) framework. Delwiche and Cooke [39] extended this hydraulic framework analytically and identified the parameter regimes in which a purely hydropassive circuit produces oscillations consistent with Fig. 1.3a. These foundational HP references locate the oscillatory regimes within the hydraulic circuit but encode no molecular information about the guard cell processes that set those parameters.

Buckley integrated guard and subsidiary cell turgor mechanics, hydroactive feedback via ATP-driven osmoregulation, and the Farquhar–Wong photosynthesis coupling into a single hydromechanical model[47]. It predicted both wrong-way and right-way responses and identified the central role of the guard–epidermal turgor pressure difference in determining g_s. Buckley and coauthors [47, 48] also demonstrated that HP processes alone cannot explain the magnitude of drought-driven stomatal closure in angiosperms, establishing that HA regulation is required and directly motivating the experimental and modelling work of this thesis[8, 9].

Chen et al. and Hills et al. [49, 50] developed OnGuard, a comprehensive kinetic model of guard cell plasma membrane and tonoplast transporters. Wang et al. [51] extended it to the leaf scale (OnGuard2), coupling guard cell ion transport to leaf water status and predicting VPD responses without explicit ABA. OnGuard2 is the principal complementary platform that resolves guard-cell electrophysiology at much greater resolution but treats ABA as a prescribed input rather than a dynamically regulated variable[52]. Chapter 3 of this thesis argues that this choice matters, because ABA autoregulation, not merely ABA levels, is necessary to explain the oscillations and hysteresis illustrated in Fig. 1.3[53]. The most recent OnGuard3e [54] adds an ecophysiology interface but retains the phenomenological ABA prescription, leaving the gap that Chapter 3 fills.

Peak and Mott [55] demonstrated that g_s responds steeply to the vapor pressure difference in the sub-stomatal cavity. Their model proposed that guard cells equilibrate to vapor in the SSC near the pore base, providing a vapor-phase mechanism for humidity sensing. This mechanism requires near-saturation in the sub-stomatal cavity, a condition that AquaDust measurements in Chapter 2 show to be violated under moderate drought[31, 32, 33]. Blatt et al. [56] have recently proposed that cell-wall water buffers guard cells against falling airspace humidity, providing a new perspective on the undersaturation debate that directly engages with the findings of Chapters 2 and 4.

Jain et al. [33] is the experimental cornerstone of Chapters 2 and 4. By simultaneously measuring ψ_{xyl} and ψ_{ssc}^{apo} in intact transpiring maize leaves, they showed a 25-fold decline in g_{oxz} beginning well above the turgor-loss point, with ψ_{ssc}^{apo} reaching -10 MPa, entirely independently of VPD and stomatal behavior. This VPD-independence directly contradicts the evaporative-demand-driven undersaturation proposed by Márquez et al. and Wong et al. [57, 58], and undermines the vapor-phase sensing framework of Peak and Mott[55, 17, 18, 51].

The work of Desai and Stroock [53] that emerged out of this thesis presents the first coupled HP–HA framework that resolves the hydraulic circuit from xylem to guard cell explicitly, couples it to a mechanistic model of guard-cell-autonomous ABA biosynthesis driven by guard cell turgor, ABA positive feedback, and ABA catabolism via A8H/CYP707A, and predicts genotype-specific transient responses in Arabidopsis genotypes. This model occupies the intersection of the hydraulic and biochemical frameworks, bridging Cowan [13] at one end and the OnGuard family at the other.

1.5 Thesis scope, organization, and contributions

The review above identifies four unresolved problems that motivate the experimental and modelling chapters of this thesis. First, no study had combined genetically suppressed hydroactive (HA) regulation with direct in-planta water potential measurement to quantify the purely passive contribution to stomatal regulation; this gap is addressed in Chapter 2. Second, no prior hydropassive–hydroactive (HP–HA) model had explicitly resolved ABA biosynthesis kinetics and linked them to both transient behaviors (wrong-way responses and oscillations; Fig. 1.3a) and steady-state behavior (hysteresis; Fig. 1.3b); this gap is addressed in Chapter 3. Third, the non-stomatal outside-xylem (OXZ) hydraulic phenotype was known only in maize, and the characterization of its importance in cotton remained only preliminary; this gap is addressed in Chapter 4. Fourth, standard soil-hydraulic parameterizations are extrapolated far beyond their calibration range during severe drought, and direct ψ_{soil} measurements down to about −30 MPa are needed to constrain them; this gap is addressed in Chapter 5.

The chapters in this thesis are organized as follows.

Chapter 2 presents a direct quantification of stomatal and non-stomatal (OXZ) physiology in wildtype and *slac1* maize across a drought trajectory using AquaDust and LI-COR 6800 gas exchange[59, 33, 60]. A purely hydropassive model fitted by Bayesian Markov chain Monte Carlo exclusively to *slac1* data is then used to infer the hydroactive contribution in wildtype, so that systematic deviations of wildtype data from HP-model predictions quantify where and how much active HA input is required[9]. The ratio g_s/g_{oxz} emerges as a mechanistically grounded water-use-efficiency phenotype that is actionable as both a screening metric and a modelling target[33].

Chapter 3 develops and validates a coupled HP–HA model incorporating guard-cell-autonomous ABA autoregulation, including a stress-driven biosynthesis term, a positive feedback amplification term, and a catabolism term via A8H/CYP707A, alongside the PYR–PP2C–OST1–SLAC1 signaling cascade. The model predicts genotype-specific damped and sustained oscillations and bistable hysteresis in Arabidopsis genotypes, establishing ABA autoregulation as the core nonlinearity underlying the dynamic behaviors illustrated in Fig. 1.3[53, 61].

Chapter 4 uses AquaDust to characterize OXZ and stomatal responses in wildtype and GhSLAC1 cotton across a range of ψ_{xyl}–VPD combinations. Cotton is found to remain stomata-dominated in sharp contrast to maize, establishing a C_3–C_4 species contrast in OXZ vulnerability and revisiting, in a direct-measurement framework, the debate on the sites of evaporation within the leaf initiated by Jarvis and Slatyer [30].

Chapter 5 implements a one-dimensional transient Richards-equation model with van Genuchten water-retention characteristics and isothermal Philip de Vries vapor conductivity, confronted against AquaSheet measurements[62, 21, 22, 63]. The chapter shows that standard constitutive parameterizations of soil fail below about −20 MPa, calling for re-parameterization in the extreme dry regime that increasing drought frequency makes agronomically relevant[24].

Chapter 6 synthesizes the four experimental and modelling themes and identifies future directions, including the development of species-specific unified HP–HA models, an ABA–OnGuard–HP hybrid with explicit A–g_s coupling[64], extensions to Ca^{2+}, reactive oxygen species (ROS), and long-distance signaling[65, 66], and a coupled soil–root–stomata SPAC model constrained jointly by AquaDust and AquaSheet[23, 59, 63].

References

[1] Jack S A Matthews and Tracy Lawson. "Climate Change and Stomatal Physiology". In: *Annual Plant Reviews online*. Major Reference Works. Aug. 2019, pp. 713–752. ISBN: 9781119312994. DOI: https://doi.org/10.1002/9781119312994.apr0667. URL: https://doi.org/10.1002/9781119312994.apr0667.

[2] Charlotte Grossiord et al. "Plant responses to rising vapor pressure deficit". In: *New Phytologist* 226 (6 June 2020), pp. 1550–1566. ISSN: 0028-646X. DOI: https://doi.org/10.1111/nph.16485. URL: https://doi.org/10.1111/nph.16485.

[3] Kimberly A Novick et al. “The impacts of rising vapour pressure deficit in natural and managed ecosystems”. In: *Plant, Cell & Environment* 47 (9 Sept. 2024), pp. 3561–3589. ISSN: 0140-7791. DOI: https://doi.org/10.1111/pce.14846. URL: https://doi.org/10.1111/pce.14846.

[4] Paul E Verslues et al. “Burning questions for a warming and changing world: 15 unknowns in plant abiotic stress”. In: *The Plant Cell* 35 (1 Jan. 2023), pp. 67–108. ISSN: 1040-4651. DOI: 10.1093/plcell/koac263. URL: https://doi.org/10.1093/plcell/koac263.

[5] Thomas N Buckley and Keith A Mott. “Modelling stomatal conductance in response to environmental factors”. In: *Plant, Cell & Environment* 36 (9 Sept. 2013), pp. 1691–1699. ISSN: 0140-7791. DOI: https://doi.org/10.1111/pce.12140. URL: https://doi.org/10.1111/pce.12140.

[6] Tracy Lawson and Michael R Blatt. “Stomatal Size, Speed, and Responsiveness Impact on Photosynthesis and Water Use Efficiency ”. In: *Plant Physiology* 164 (4 Apr. 2014), pp. 1556–1570. ISSN: 0032-0889. DOI: 10.1104/pp.114.237107. URL: https://doi.org/10.1104/pp.114.237107.

[7] G D Farquhar et al. “Carbon Isotope Fractionation and Plant Water-Use Efficiency”. In: *Stable Isotopes in Ecological Research*. Ed. by P W Rundel, J R Ehleringer, and K A Nagy. Springer New York, 1989, pp. 21–40. ISBN: 978-1-4612-3498-2.

[8] Thomas N Buckley. “Modeling Stomatal Conductance.” In: *Plant physiology* 174 (2 June 2017), pp. 572–582. ISSN: 1532-2548 (Electronic). DOI: 10.1104/pp.16.01772.

[9] Thomas N Buckley. “How do stomata respond to water status?” In: *New Phytologist* 224 (1 Oct. 2019). https://doi.org/10.1111/nph.15899, pp. 21–36. ISSN: 0028-646X. DOI: https://doi.org/10.1111/nph.15899. URL: https://doi.org/10.1111/nph.15899.

[10] Belinda E Medlyn et al. “Reconciling the optimal and empirical approaches to modelling stomatal conductance”. In: *Global Change Biology* 17 (6 June 2011), pp. 2134–2144. ISSN: 1354-1013. DOI: https://doi.org/10.1111/j.1365-2486.2010.02375.x. URL: https://doi.org/10.1111/j.1365-2486.2010.02375.x.

[11] Alistair M Hetherington and F Ian Woodward. “The role of stomata in sensing and driving environmental change”. In: *Nature* 424 (6951 2003), pp. 901–908. ISSN: 1476-4687. DOI: 10.1038/nature01843. URL: https://doi.org/10.1038/nature01843.

[12] Park S Nobel. “Chapter 8 - Leaves and Fluxes”. In: *Physicochemical and Environmental Plant Physiology (Fourth Edition)*. Academic Press, 2009, pp. 364–437. ISBN: 978-0-12-374143-1. DOI: https://doi.org/10.1016/B978-0-12-374143-1.00008-9. URL: https://www.sciencedirect.com/science/article/pii/B9780123741431000089.

[13] I R Cowan. “Oscillations in stomatal conductance and plant functioning associated with stomatal conductance: Observations and a model”. In: *Planta* 106 (3 1972), pp. 185–219. ISSN: 1432-2048. DOI: 10.1007/BF00388098. URL: https://doi.org/10.1007/BF00388098.

[14] P G Jarvis. “The interpretation of the variations in leaf water potential and stomatal conductance found in canopies in the field”. In: *Philosophical Transactions of the Royal Society of London. B, Biological Sciences* 273 (927 Feb.

1976), pp. 593–610. ISSN: 0080-4622. DOI: 10.1098/rstb.1976.0035. URL: https://doi.org/10.1098/rstb.1976.0035.

[15] I R Cowan and G D Farquhar. "Stomatal function in relation to leaf metabolism and environment". In: *Symposia of the Society for Experimental Biology* 31 (1977), pp. 471–505. ISSN: 0081-1386. URL: https://www.ncbi.nlm.nih.gov/pubmed/756635.

[16] P. Gaastra. "Photosynthesis of crop plants as influenced by light, carbon dioxide, temperature, and stomatal diffusion resistance". In: (Dec. 1959). DOI: 10.18174/183268. URL: https://research.wur.nl/en/publications/photosynthesis-of-crop-plants-as-influenced-by-light-carbon-dioxi/.

[17] Fulton E Rockwell. "Shorting the metaphorical circuit: vascular partitioning and stomatal patchiness can create apparent unsaturation and CO2 gradient inversion in the Ohmic analogy for leaf gas exchange". In: *New Phytologist* 244 (5 Dec. 2024), pp. 1812–1823. ISSN: 0028-646X. DOI: https://doi.org/10.1111/nph.20106. URL: https://doi.org/10.1111/nph.20106.

[18] Fulton E Rockwell et al. "Extreme undersaturation in the intercellular airspace of leaves: a failure of Gaastra or Ohm?" In: *Annals of Botany* 130 (3 Sept. 2022), pp. 301–316. ISSN: 0305-7364. DOI: 10.1093/aob/mcac094. URL: https://doi.org/10.1093/aob/mcac094.

[19] J S Sperry et al. "Limitation of plant water use by rhizosphere and xylem conductance: results from a model". In: *Plant, Cell & Environment* 21 (4 Apr. 1998), pp. 347–359. ISSN: 0140-7791. DOI: https://doi.org/10.1046/j.1365-3040.1998.00287.x. URL: https://doi.org/10.1046/j.1365-3040.1998.00287.x.

[20] J S Sperry et al. "Water deficits and hydraulic limits to leaf water supply". In: *Plant, Cell & Environment* 25 (2 Feb. 2002), pp. 251–263. ISSN: 0140-7791. DOI: https://doi.org/10.1046/j.0016-8025.2001.00799.x. URL: https://doi.org/10.1046/j.0016-8025.2001.00799.x.

[21] L A Richards. "Capillary conduction of liquids through porous mediums". In: *Physics* 1 (5 Nov. 1931), pp. 318–333. ISSN: 0148-6349. DOI: 10.1063/1.1745010. URL: https://doi.org/10.1063/1.1745010.

[22] M Th van Genuchten. "A closed-form equation for predicting the hydraulic conductivity of unsaturated soils". In: *Soil Science Society of America Journal. Soil Science Society of America* 44 (5 1980), pp. 892–898. ISSN: 0361-5995. DOI: 10.2136/sssaj1980.03615995004400050002x. URL: http://dx.doi.org/10.2136/sssaj1980.03615995004400050002x.

[23] Andrea Carminati and Mathieu Javaux. "Soil Rather Than Xylem Vulnerability Controls Stomatal Response to Drought". In: *Trends in Plant Science* 25 (9 2020), pp. 868–880. ISSN: 1360-1385. DOI: https://doi.org/10.1016/j.tplants.2020.04.003. URL: https://www.sciencedirect.com/science/article/pii/S1360138520301187.

[24] Sara Di Bert et al. "Early signals of water limitations begin at the root–soil interface: linking rhizosphere drying to water uptake decline". In: *New Phytologist* 249 (6 Mar. 2026), pp. 2909–2918. ISSN: 0028-646X. DOI: https://doi.org/10.1111/nph.70879. URL: https://doi.org/10.1111/nph.70879.

[25] Anju Manandhar et al. "Dynamic soil hydraulic resistance regulates stomata". In: *New Phytologist* 244 (1 Oct. 2024), pp. 147–158. ISSN: 0028-646X.

DOI: https://doi.org/10.1111/nph.20020. URL: https://doi.org/10.1111/nph.20020.

[26] Peter J Franks. "Use of the pressure probe in studies of stomatal function". In: *Journal of Experimental Botany* 54 (387 June 2003), pp. 1495–1504. ISSN: 0022-0957. DOI: 10.1093/jxb/erg162. URL: https://doi.org/10.1093/jxb/erg162.

[27] P J Franks, 2 I R Cowan 1, and G D Farquhar. "A study of stomatal mechanics using the cell pressure probe". In: *Plant, Cell & Environment* 21 (1 1998), pp. 94–100. DOI: 10.1046/j.1365-3040.1998.00248.x. URL: https://onlinelibrary.wiley.com/doi/abs/10.1046/j.1365-3040.1998.00248.x.

[28] Peter J Franks and Graham D Farquhar. "The Mechanical Diversity of Stomata and Its Significance in Gas-Exchange Control". In: *Plant Physiology* 143 (1 Jan. 2007), pp. 78–87. ISSN: 0032-0889. DOI: 10.1104/pp.106.089367. URL: https://doi.org/10.1104/pp.106.089367.

[29] K Raschke. "Stomatal Action". In: *Annual Review of Plant Physiology* 26 (1 June 1975), pp. 309–340. ISSN: 0066-4294. DOI: 10.1146/ANNUREV.PP.26.060175.001521.

[30] P G Jarvis and R O Slatyer. "The role of the mesophyll cell wall in leaf transpiration". In: *Planta* 90 (4 1970), pp. 303–322. ISSN: 1432-2048. DOI: 10.1007/BF00386383. URL: https://doi.org/10.1007/BF00386383.

[31] Piyush Jain et al. "New approaches to dissect leaf hydraulics reveal large gradients in living tissues of tomato leaves". In: *New Phytologist* 242 (2

Apr. 2024), pp. 453–465. ISSN: 0028-646X. DOI: https://doi.org/10.1111/nph.19585. URL: https://doi.org/10.1111/nph.19585.

[32] Piyush Jain et al. "Localized measurements of water potential reveal large loss of conductance in living tissues of maize leaves". In: *Plant Physiology* 194 (4 Apr. 2024), pp. 2288–2300. ISSN: 0032-0889. DOI: 10.1093/plphys/kiad679. URL: https://doi.org/10.1093/plphys/kiad679.

[33] Piyush Jain et al. "Loss of conductance between mesophyll symplasm and intercellular air spaces explains nonstomatal control of transpiration". In: *Proceedings of the National Academy of Sciences* 122 (47 Nov. 2025). doi: 10.1073/pnas.2504862122, e2504862122. DOI: 10.1073/pnas.2504862122. URL: https://doi.org/10.1073/pnas.2504862122.

[34] Sean R Cutler et al. "Abscisic acid: emergence of a core signaling network." In: *Annual review of plant biology* 61 (2010), pp. 651–679. ISSN: 1545-2123 (Electronic). DOI: 10.1146/annurev-arplant-042809-112122.

[35] Po-Kai Hsu et al. "Signaling mechanisms in abscisic acid-mediated stomatal closure". In: *The Plant Journal* 105 (2 Jan. 2021), pp. 307–321. ISSN: 0960-7412. DOI: https://doi.org/10.1111/tpj.15067. URL: https://doi.org/10.1111/tpj.15067.

[36] Shintaro Munemasa et al. "Mechanisms of abscisic acid-mediated control of stomatal aperture." In: *Current opinion in plant biology* 28 (Dec. 2015), pp. 154–162. ISSN: 1879-0356 (Electronic). DOI: 10.1016/j.pbi.2015.10.010.

[37] Julian I. Schroeder et al. “Guard cell signal transduction”. In: *Annual Review of Plant Biology* 52 (Volume 52, 2001 June 2001), pp. 627–658. ISSN: 15435008. DOI: 10.1146/ANNUREV.ARPLANT.52.1.627/CITE/REFWORKS. URL: https://www.annualreviews.org/content/journals/10.1146/annurev.arplant.52.1.627.

[38] Joshua J Weiner et al. “Structural and functional insights into core ABA signaling.” In: *Current opinion in plant biology* 13 (5 Oct. 2010), pp. 495–502. ISSN: 1879-0356 (Electronic). DOI: 10.1016/j.pbi.2010.09.007.

[39] Michael J Delwiche and J Robert Cooke. “An analytical model of the hydraulic aspects of stomatal dynamics”. In: *Journal of theoretical biology* 69 (1 1977), pp. 113–141. ISSN: 0022-5193.

[40] Graham D Farquhar and Ian R Cowan. “Oscillations in Stomatal Conductance: The Influence of Environmental Gain”. In: *Plant Physiology* 54 (5 Nov. 1974), pp. 769–772. ISSN: 0032-0889. DOI: 10.1104/pp.54.5.769. URL: https://doi.org/10.1104/pp.54.5.769.

[41] Yong Ding, Michael Fromm, and Zoya Avramova. “Multiple exposures to drought’train’transcriptional responses in Arabidopsis”. In: *Nature communications* 3 (1 2012), p. 740. ISSN: 2041-1723.

[42] Yohei Takahashi and Toshinori Kinoshita. “Stomatal function has an element of hysteresis”. In: *New Phytologist* 205 (2 Jan. 2015), pp. 455–457. ISSN: 0028-646X. DOI: https://doi.org/10.1111/nph.13149. URL: https://doi.org/10.1111/nph.13149.

[43] Laetitia Virlouvet and Michael Fromm. “Physiological and transcriptional memory in guard cells during repetitive dehydration stress”. In: *New Phytologist* 205 (2 Jan. 2015), pp. 596–607. ISSN: 0028-646X. DOI: https://

doi.org/10.1111/nph.13080. URL: https://doi.org/10.1111/nph.13080.

[44] J Timothy Ball, Ian E Woodrow, and Joseph A Berry. "A Model Predicting Stomatal Conductance and its Contribution to the Control of Photosynthesis under Different Environmental Conditions". In: *Progress in Photosynthesis Research: Volume 4 Proceedings of the VIIth International Congress on Photosynthesis Providence, Rhode Island, USA, August 10–15, 1986*. Ed. by J Biggins. Springer Netherlands, 1987, pp. 221–224. ISBN: 978-94-017-0519-6. DOI: 10.1007/978-94-017-0519-6_48. URL: https://doi.org/10.1007/978-94-017-0519-6_48.

[45] R Leuning. "A critical appraisal of a combined stomatal-photosynthesis model for C3 plants". In: *Plant, Cell & Environment* 18 (4 Apr. 1995), pp. 339–355. ISSN: 0140-7791. DOI: https://doi.org/10.1111/j.1365-3040.1995.tb00370.x. URL: https://doi.org/10.1111/j.1365-3040.1995.tb00370.x.

[46] Gabriel Katul et al. "A stomatal optimization theory to describe the effects of atmospheric CO2 on leaf photosynthesis and transpiration". In: *Annals of Botany* 105 (3 Mar. 2010), pp. 431–442. ISSN: 0305-7364. DOI: 10.1093/aob/mcp292. URL: https://doi.org/10.1093/aob/mcp292.

[47] T N Buckley, K A Mott, and G D Farquhar. "A hydromechanical and biochemical model of stomatal conductance". In: *Plant, Cell & Environment* 26 (10 Oct. 2003), pp. 1767–1785. ISSN: 0140-7791. DOI: https://doi.org/10.1046/j.1365-3040.2003.01094.x. URL: https://doi.org/10.1046/j.1365-3040.2003.01094.x.

[48] T N Buckley and K A Mott. "Dynamics of stomatal water relations during the humidity response: implications of two hypothetical mechanisms". In:

Plant, Cell & Environment 25 (3 Mar. 2002), pp. 407–419. ISSN: 0140-7791. DOI: https://doi.org/10.1046/j.0016-8025.2001.00820.x. URL: https://doi.org/10.1046/j.0016-8025.2001.00820.x.

[49] Zhong-Hua Chen et al. "Systems Dynamic Modeling of the Stomatal Guard Cell Predicts Emergent Behaviors in Transport, Signaling, and Volume Control ". In: *Plant Physiology* 159 (3 July 2012), pp. 1235–1251. ISSN: 0032-0889. DOI: 10.1104/pp.112.197350. URL: https://doi.org/10.1104/pp.112.197350.

[50] Adrian Hills et al. "OnGuard, a computational platform for quantitative kinetic modeling of guard cell physiology". In: *Plant physiology* 159 (3 July 2012), pp. 1026–1042. ISSN: 1532-2548. DOI: 10.1104/pp.112.197244. URL: https://pubmed.ncbi.nlm.nih.gov/22635116%20https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3387691/.

[51] Yizhou Wang et al. "Unexpected Connections between Humidity and Ion Transport Discovered Using a Model to Bridge Guard Cell-to-Leaf Scales". In: *The Plant Cell* 29 (11 Nov. 2017), pp. 2921–2939. ISSN: 1040-4651. DOI: 10.1105/tpc.17.00694. URL: https://doi.org/10.1105/tpc.17.00694.

[52] Mareike Jezek et al. "A constraint–relaxation–recovery mechanism for stomatal dynamics". In: *Plant, Cell & Environment* 42 (8 Aug. 2019). https://doi.org/10.1111/pce.13568, pp. 2399–2410. ISSN: 0140-7791. DOI: https://doi.org/10.1111/pce.13568. URL: https://doi.org/10.1111/pce.13568.

[53] Sahil Desai and Abraham Stroock. "Abscisic acid-mediated water stress regulation can mechanistically explain oscillations and water stress memory in stomatal conductance". In: *bioRxiv* (Jan. 2025), p. 2025.12.26.696581.

DOI: 10.64898/2025.12.26.696581. URL: http://biorxiv.org/content/early/2025/12/29/2025.12.26.696581.abstract.

[54] Thanh-Hao Nguyen et al. "OnGuard3e: A predictive, ecophysiology-ready tool for gas exchange and photosynthesis research". In: *Plant, Cell & Environment* 46 (11 Nov. 2023), pp. 3644–3658. ISSN: 0140-7791. DOI: https://doi.org/10.1111/pce.14674. URL: https://doi.org/10.1111/pce.14674.

[55] David Peak and Keith A Mott. "A new, vapour-phase mechanism for stomatal responses to humidity and temperature". In: *Plant, Cell & Environment* 34 (1 Jan. 2011), pp. 162–178. ISSN: 0140-7791. DOI: https://doi.org/10.1111/j.1365-3040.2010.02234.x. URL: https://doi.org/10.1111/j.1365-3040.2010.02234.x.

[56] Michael R Blatt et al. "Cell wall water shields stomata against falling leaf airspace humidity". In: *New Phytologist* 250 (2 Apr. 2026), pp. 861–872. ISSN: 0028-646X. DOI: https://doi.org/10.1111/nph.70998. URL: https://doi.org/10.1111/nph.70998.

[57] Diego A Márquez et al. "Mesophyll airspace unsaturation drives C4 plant success under vapor pressure deficit stress". In: *Proceedings of the National Academy of Sciences* 121 (39 Sept. 2024). doi: 10.1073/pnas.2402233121, e2402233121. DOI: 10.1073/pnas.2402233121. URL: https://doi.org/10.1073/pnas.2402233121.

[58] Suan Chin Wong et al. "Humidity gradients in the air spaces of leaves". In: *Nature Plants* 8 (8 2022), pp. 971–978. ISSN: 2055-0278. DOI: 10.1038/s41477-022-01202-1. URL: https://doi.org/10.1038/s41477-022-01202-1.

[59] Piyush Jain et al. "A minimally disruptive method for measuring water potential in planta using hydrogel nanoreporters". In: *Proceedings of the National Academy of Sciences* 118 (23 June 2021). doi: 10.1073/pnas.2008276118, e2008276118. DOI: 10 . 1073 / pnas . 2008276118. URL: https : / / doi . org / 10 . 1073 / pnas . 2008276118.

[60] Robert J Twohey et al. "Investigating the role of stomatal dynamics on agronomic traits using a slac1-2 Zea mays mutant". In: *bioRxiv* (Jan. 2025), p. 2025.01.21.634166. DOI: 10.1101/2025.01.21.634166. URL: http://biorxiv.org/content/early/2025/01/24/2025.01.21.634166.abstract.

[61] Ebe Merilo et al. "Stomatal VPD Response: There Is More to the Story Than ABA ". In: *Plant Physiology* 176 (1 Jan. 2018), pp. 851–864. ISSN: 0032-0889. DOI: 10.1104/pp.17.00912. URL: https://doi.org/10.1104/pp.17.00912.

[62] J R Philip and D A De Vries. "Moisture movement in porous materials under temperature gradients". In: *Transactions* 38 (2 1957), p. 222. ISSN: 0002-8606. DOI: 10.1029/tr038i002p00222. URL: http://dx.doi.org/10.1029/tr038i002p00222.

[63] I-Feng Wu et al. "An Optical Reporting Patch for Spatiotemporal Measurements of Water Activity in Complex Environments". In: *Analytical Chemistry* 98 (16 Apr. 2026). doi: 10.1021/acs.analchem.5c08117, pp. 11865–11877. ISSN: 0003-2700. DOI: 10 . 1021 / acs . analchem . 5c08117. URL: https : / / doi . org / 10 . 1021 / acs . analchem . 5c08117.

[64] Mareike Jezek et al. “Guard cell endomembrane Ca2+-ATPases underpin a ’carbon memory’ of photosynthetic assimilation that impacts on water-use efficiency”. In: *Nature Plants* 7 (9 2021), pp. 1301–1313. ISSN: 2055-026X. DOI: 10.1038/s41477-021-00966-2. URL: http://dx.doi.org/10.1038/s41477-021-00966-2%20https://www.ncbi.nlm.nih.gov/pubmed/34326530.

[65] Vesna Bacheva et al. “A unified framework for hydromechanical signaling can explain transmission of local and long-distance signals in plants”. In: *Proceedings of the National Academy of Sciences* 122 (17 Apr. 2025). doi: 10.1073/pnas.2422692122, e2422692122. DOI: 10.1073/pnas.2422692122. URL: https://doi.org/10.1073/pnas.2422692122.

[66] Annalisa Bellandi et al. “Diffusion and bulk flow of amino acids mediate calcium waves in plants”. In: *Science Advances* 8 (42 June 2022). doi: 10.1126/sciadv.abo6693, eabo6693. DOI: 10.1126/sciadv.abo6693. URL: https://doi.org/10.1126/sciadv.abo6693.